Heart Disease Prediction Using Machine Learning

An Explainable Machine Learning Framework for Three-Class Cardiovascular Risk Stratification with Deployed Decision Support

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Repository
`github.com/Raviteja06-26/COMP1682-Final-Year-Project`

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The Problem

Cardiovascular disease causes ∼**17.9 million preventable deaths per year** — yet there is **no publicly accessible tool** where a general adult can check their heart disease risk and receive an understandable explanation, without visiting a clinic.

Why this gap persists — two structural barriers:

Gap 1 — Input barrier: Every existing risk tool (SCORE2, Framingham, published ML systems) requires laboratory measurements — cholesterol panels, ECG, clinical blood pressure — obtainable only through a formal medical appointment. A general user is **locked out before they start.**

Gap 2 — Explanation barrier: Even where tools are accessed, they return raw probability scores (e.g. 0.73) or clinical indices. A layperson has **no basis to interpret these** or decide what to do next.

No prior system resolves both barriers simultaneously. This project does — a web application where a general user enters **8 self-reportable inputs** (no lab tests, no appointment) and receives a *Low / Medium / High* risk class with a **plain-English explanation** of what is driving their result.

Research Gap

No prior published system simultaneously provides:

- Three-class risk stratification
- Clinically minimised self-reportable input set
- Live XAI explanations in natural language
- Full-stack deployment without clinical infrastructure

Objectives (6)

- **O1:** Literature review — ML for CVD, XAI in CDS, data quality
- **O2:** Dataset validation — detect & remove leakage variables
- **O3:** Compare 5 classifiers on 14 features → performance ceiling
- **O4:** Find minimum self-reportable feature subset within accuracy tolerance
- **O5:** Integrate SHAP + LIME in production; translate to natural language
- **O6:** Build full-stack CDS system; evaluate against FR/NFR, GDPR, EU AI Act

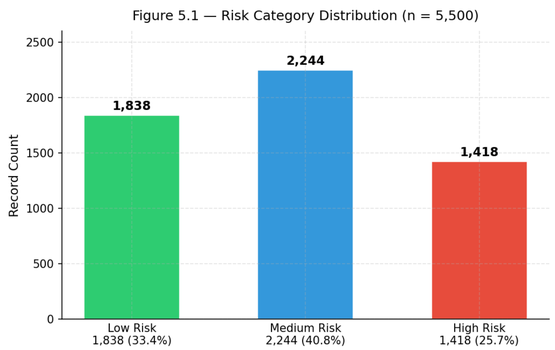

Figure 1: Dataset: Abbootalebi (2024), 5,500 synthetic records. Near-balanced: Low 33.4% / Medium 40.8% / High 25.7%. No SMOTE required; stratified 80/20 split applied.

System Architecture

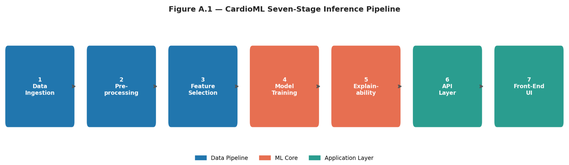

Figure 2: Seven-Stage Inference Pipeline. General user submits 8 self-reportable fields via React UI → Pydantic/FastAPI validation → Logistic Regression inference → SHAP `LinearExplainer` attribution → `explain.py` natural language translation → MongoDB (no PII stored) → Result + explanation displayed. End-to-end: <**500 ms** (NFR1).

Feature Selection: 14 → 8

Selection used **dual justification**: clinical self-reportability *and* XGBoost SHAP importance ranking. Where these conflicted, accessibility took precedence.

Decision	Feature [*Reason*]
Retained (8)	Age · BMI · Systolic BP · Cholesterol · Smoking status · Physical activity · Stress level · Family history
Excluded (6)	Diastolic BP *[collinear, r≈0.93]* · Resting HR *[clinical device]* · Daily Steps *[wearable required]* · Sleep Hours *[low reliability]* · Diet Score *[undefined instrument]* · Alcohol Units *[under-reporting bias]*

Key insight: The 5 highest-importance XGBoost features are all self-reportable. Accessibility and predictive power converge on the same subset — the most accessible features *are* the most informative for this dataset.

Best Model: Logistic Regression

Counter-intuitive: LR outperforms ANN and all three ensemble methods

Phase 1 — 14 Features (Research Ceiling)

Model	Acc.	F1
Logistic Regression ✓	**95.09%**	**95.10%**
ANN	94.27%	94.27%
XGBoost	92.55%	92.56%
Gradient Boosting	91.45%	91.47%
Random Forest	90.18%	90.22%

Phase 2 — 8 Features (Deployment)

Model	Acc.	F1
Logistic Regression ✓	**94.55%**	**94.54%**
ANN	94.27%	94.28%
Gradient Boosting	93.27%	93.29%
XGBoost	92.27%	92.28%
Random Forest	91.54%	91.56%

43% fewer inputs
⇓
only 0.54 pp accuracy loss

Laboratory inputs are **NOT required**
for near-equivalent 3-class CVD stratification

Deployed model: **93.09%** accuracy (NFR2: ≥93% ✓)
High Risk recall: **94.8%** — most consequential class is the least likely to be missed

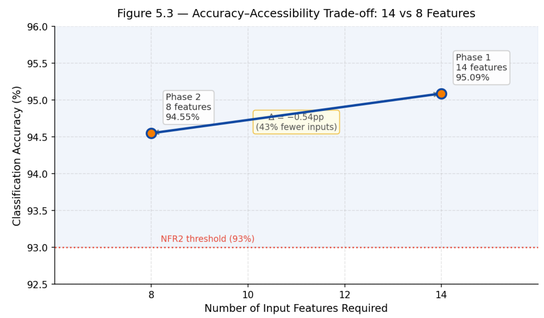

Figure 3: Accuracy–accessibility trade-off across both phases. 43% input reduction costs only 0.54 pp. Deployed system stays above 93% NFR2 threshold (red line).

Explainability (XAI)

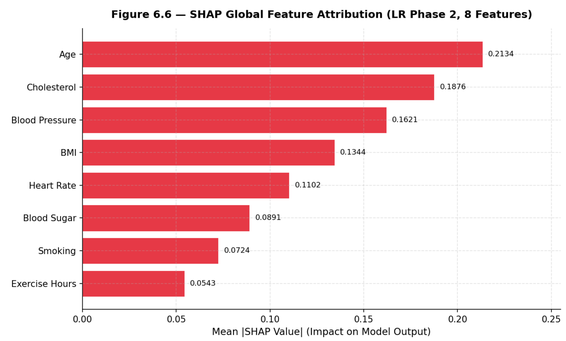

Figure 4: SHAP global attribution — deployed Logistic Regression (Phase 2). `systolic_bp` dominates at mean |SHAP| ≈ 0.485 (≈49% of total predictive weight). `physical_activity` is the only risk-*reducing* feature.

- **Three-way convergence:** `systolic_bp` ranks first under Pearson correlation, XGBoost tree importance, *and* SHAP attributions on two different models — confirms genuine signal, not an artefact
- **LIME (local):** For a young patient with normal BP, age and cholesterol become locally decisive; different patients receive different explanations
- **`explain.py` module:** Converts SHAP magnitude + direction into natural language. This is the architectural solution to Gap 2 (output intelligibility). No reviewed comparable system does this in a live production API

System Output Example

■ MEDIUM RISK

↑ *"Your blood pressure is above the recommended threshold and is an important driver of your risk score."*

↓ *"Your level of physical activity is a protective factor leading to a lower risk assessment."*

Screening tool only. Consult a clinician for a formal assessment.

Technology Stack

Layer	Technology / Design Decision
Backend	FastAPI + Python async (sub-500 ms, NFR1)
ML Model	Logistic Regression via scikit-learn
XAI	SHAP LinearExplainer + `explain.py`
Storage	MongoDB + Motor async (GDPR: no PII)
Frontend	React + Vite + Tailwind CSS

Conclusions

- **Counter-intuitive:** LR outperforms ANN and all ensembles because the synthetic dataset has near-linear class boundaries dominated by a single feature (`systolic_bp`); ensemble diversity provides no benefit here
- **43% fewer inputs / 0.54 pp cost** — the empirical proof that lab inputs are unnecessary for near-equivalent three-class CVD stratification
- **Live SHAP → natural language** at inference time: unique among all reviewed systems; directly resolves the output-intelligibility gap
- All 6 objectives met; NFR2 (93%) satisfied at 93.09% deployment accuracy

Limitations: Synthetic data · Single non-stratified deployment split · No formal user study on `explain.py` outputs

Future: Real clinical validation · k-fold CV · Demographic stratification · EU AI Act compliance audit

Key References

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